Lukas José Ferrer

ML & AI SYSTEMS ENGINEER

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LinkedIn

ML engineer building clinical AI, self-hosted inference, and developer tools. Shipped systems processing **100M+ medical-record pages**; creator of **WAR (Work, Audit, Refine)**, helping engineers turn design decisions into implemented, audited code.

FEATURED OPEN SOURCE

WAR (Work, Audit, Refine)

Creator & Maintainer | June 2026 - Present

- An open-source package for Claude Code with a collection of skills and hooks ranging from plan drafting, to advanced memory management, to user message readability (mitigating 'Claudanese') and code reliability.
- Built a planning skill that turns open-ended goals into evidence-backed plans with explicit design decisions, acceptance criteria, and implementation latitude (vastly superior to /grill-me).
- Created an execution engine that follows the agreed plan, resolves in-scope implementation ambiguity, and asks the engineer when the design must change.
- Independent, read-only auditors are first-class subagents, driving in-scope bug fixes to converge to a stable solution, freeing engineers to focus on design decisions and code review.

Explore WAR

Watch demo

EXPERIENCE

Lead ML & DevOps Engineer | Sequoia Port Inc.

Mar 2021 - Present | Remote / Los Angeles, CA

- **Cut AI Activation processing costs from ~$0.031 to <$0.01/page** by migrating to self-hosted Qwen 3.6 (27B). At **500K+ pages/month**, this represents **~$125K+ in annualized processing savings**.
- Architected the core medical-record ingestion, segmentation, and indexing pipeline serving Fortune 100 healthcare customers: **100M+ pages processed** and **5M+ pages/month**, separate from AI Activation.
- Built clinical AI products that reduced insurance claim review time by **65%** and specialist escalations by **40%**; automated segmentation of 1,000+ page cases for physician review in under an hour.
- Improved clinical retrieval from **87% to 95% Recall@5** using a domain ontology and graph database; delivered case summaries with validated source citations and PHI protection.
- **Reduced ECS compute costs by 80%** through ARM64 optimization and Spot capacity with automatic on-demand recovery.
- Built automated and on-demand evaluation pipelines to verify that NLP model migrations (including Claude to Qwen) meet or exceed deployed-system quality, using regression checks, blinded comparisons, and clinical-error analysis.

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Production AI / ML infrastructure / Agent orchestration

EXPERIENCE CONTINUED

Sequoia Port Inc. | Production quality & reliability

- Designed idempotent processing and durable persistence that preserve human reviewers' edits during AI Activation, keeping medical-record updates consistent across concurrent processing, retries, and reruns.
- Built resilient model-serving infrastructure with asynchronous execution, hybrid local/cloud GPU capacity, durable recovery, and automatic fallback to keep clinical workflows running.

Red Team (Contract) | OpenAI

Jan 2024 - Jun 2024 | Remote

- Red-teamed GPT-4o for calibrated uncertainty in clinical contexts. Designed synthetic-condition evaluations that exposed plausible but nonexistent ICD-10 codes and tested ontology-grounded retrieval.

Regulatory Software Engineer | Neuralink

Oct 2020 - Jan 2021 | Fremont, CA

- Supported FDA regulatory preparation for a Class III implantable brain-computer interface; assessed design-control gaps and proposed remediation aligned with medical-device requirements.

Research Programmer | USC Information Sciences Institute

May 2018 - Oct 2020 | Marina del Rey, CA

- Integrated research Transformer architectures into a neural machine translation training library; trained thousands of models and improved translation quality across low-resource languages.
- Built an ETL pipeline ingesting **100,000+ news articles/day** for the SAGE geopolitical forecasting platform, improving relevant-document retrieval by **73% (Recall@5)**.

TECHNICAL & PROFESSIONAL SKILLS

AI & ML: PyTorch, Transformers, multimodal models, LoRA, distillation, Qwen, Anthropic/OpenAI APIs, RAG, pgvector, FAISS, citation validation.

Evaluation: LLM red teaming, regression benchmarking, blinded model comparisons, statistical testing.

Infrastructure: AWS CDK, Lambda, Step Functions, ECS/Fargate, ECS Anywhere, EC2, SQS/SNS, S3, DynamoDB, PostgreSQL, Docker, Linux, GPU serving, CI/CD.

Development: Python, TypeScript/JavaScript, SQL, Bash, React, Ruby on Rails, Claude Code, MCP, multi-agent orchestration, Git worktrees, SQLite FTS5.

Domain: Clinical document processing, PHI protection, HIPAA-oriented pipelines, differential privacy (PATE), medical-device regulatory workflows.

Product & engineering: Requirements discovery, technical planning, design reviews, technical writing, risk assessment, and translating clinical workflows into measurable product goals.

EDUCATION

M.S. Biomedical Engineering | USC Viterbi School of Engineering | Aug 2017

B.S. Biomedical Engineering | USC Viterbi School of Engineering | May 2016